

4D Facial Capture Pipeline Incorporating Progressive Retopology Approach

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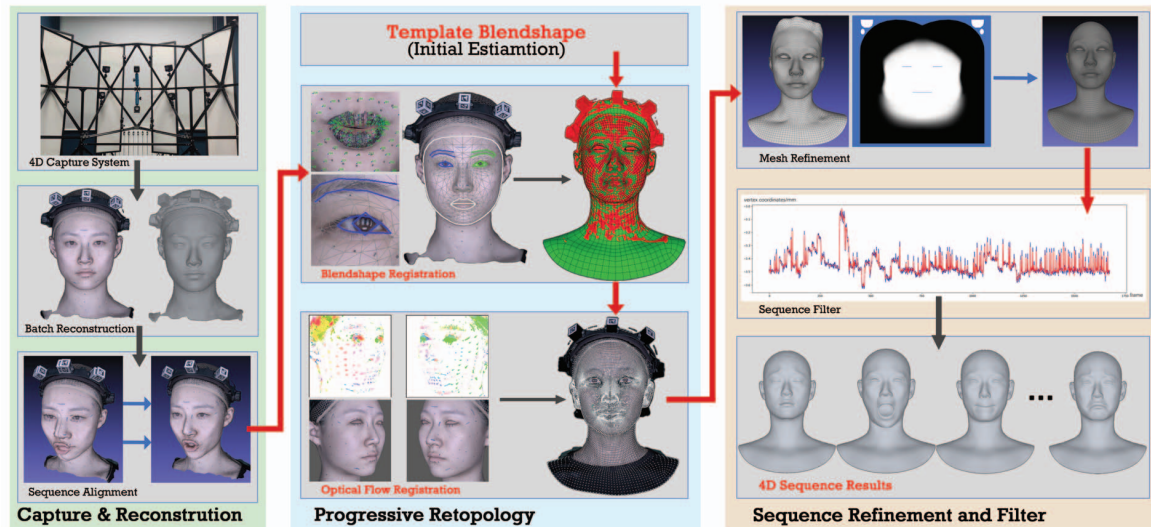


Figure 1: Overview of 4D facial capture pipeline with three stages: capture and reconstruction, progressive retopology, sequence refinement and filter.

ABSTRACT

The pipeline for creating high-fidelity facial models often utilizes multi-view stereo techniques for reconstruction. However, the subsequent step of retopology often involves intricate manual work, limiting the extension of facial capture systems towards 4D acquisition. This paper proposes a facial 4D capture pipeline based on high-speed cameras. We employ standard multi-view stereo techniques for 3D reconstruction. Non-linear deformations of facial expressions are decoupled from rigid movements of the skull using QR code markers. Additionally, a progressive automated retopology approach is introduced for batch processing. Results demonstrate that our system can capture continuous facial motion sequences with detailed 3D models.

Index Terms: Computing methodologies—Computer graphics—Animation—Motion capture; Computing methodologies—Artificial intelligence—Computer vision—Image and video acquisition

1 INTRODUCTION

Creating high-fidelity digital faces is not only highly sought after in the film and gaming industries but also holds significant importance in AR/VR for remote interactions and virtual assistants. The mainstream approach in professional studios is still to choose high-resolution multi-view images as input to ensure the highest possible fidelity and surface coverage in controlled settings. However, the current workflow only captures the extreme expressions of actors, resulting in digital assets lacking personalized facial motion information.

The typical approaches used in production and the current mainstream facial datasets include only 3D scans [5], employing non-

rigid registration to deform 3D facial models to each scan, ensuring consistent mesh topology. The registration process often requires manual landmarking tasks to assist in correspondences and parameter adjustments for accurate fitting. In a production environment, the creation of a single character typically requires up to a week for completion. The other side, existing 4D databases [2, 3] for high-resolution models rely on 3DMM and non-rigid registration methods to align facial models. This process involves only geometric feature constraints during non-rigid registration, resulting in aligned outcomes lacking semantic coherence. Furthermore, due to the stochastic nature of the non-rigid registration process, the output 4D model sequences lack temporal coherence.

Our pipeline aims to capture and process a stable facial 4D model sequence. As illustrated in Fig. 1, we utilize a custom-built matrix of high-speed industrial cameras and standard multi-view stereo techniques to reconstruct the initial high-precision model. The pipeline is achieved through alignment, registration, and refinement processes. The resulting output can be utilized in various scenarios such as facial animation generation, automatic rigging, and deep learning training.

2 4D FACIAL CAPTURE PIPELINE

2.1 Capture and reconstruction

The capture system employs 14 RGB industrial cameras with global shutters, covering a facial area of 210 degrees. The camera array consists of 6 cameras with 12.4MP resolution and 8 cameras with 3.2MP resolution. To achieve a capture rate of 60Hz, the cameras are connected to the workstation PCs via PCIe capture cards. Strict time synchronization of shutters is ensured through hard triggering of the industrial cameras. Additionally, the uniform illumination source is composed of LED panels with excellent color rendering properties. The LED lights have a sufficiently high frequency to avoid imaging flicker at low shutter times. The distributed video capture system is connected to different workstation computers via 5 PCIe cards, and the raw data we capture is approximately 6GB/s.

We employed a standard multi-view stereo pipeline to recon-

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struct accurate 3D geometry. Dense point clouds were reconstructed through deep learning-based feature detection and depth map estimation. A high-precision scanned model with 4K high-definition textures was obtained using Poisson reconstruction. However, at this point, the reconstructed model sequence contains two types of motion. The vertex motion of face includes non-linear deformations caused by facial muscles and skin, as well as global displacement introduced by the skull. Macroscopic viewing may experience random head jitter. We utilized a headgear with QR codes to represent the motion of the skull, as shown in Fig. 2. The model sequence is rendered through virtual cameras, and the corners of the QR codes are tracked and detected. The pixel values of the corners are traced to acquire three-dimensional positions through ray tracing and calculate the alignment transformation matrix. Compared to landmark-based Procrustes analysis alignment methods in related works, our approach has an absolute accuracy advantage.



Figure 2: (a) The headgear with cube of QR code. (b) The ray tracing of the QR code corners.

2.2 Progressive Automatic Retopology

To achieve registration for thousands of models in one sequence, we propose a highly robust and automated process. For any target character, manual registration is only required for one neutral expression model that includes facial texture. In the blendshape registration step, we first generate a set of standard blendshapes based on the neutral model using the deformation transfer [4] method. For the model of each frame, 2D landmarks are detected using rendering images through mediapipe [1]. The 3D landmarks computed through back-projection are used to constrain the blendshape, obtaining the 3D face initial estimation. Subsequently, non-rigid Iterative Closest Point method is employed to achieve a more accurate vertex-level fitting of dense point clouds. Local deformation smoothness constraints are also applied during the non-rigid registration process.

However, the distribution of triangles in the 3D model should have reasonable semantic coherence. The neutral template model for each character involves manual manipulation by artists, and its topological wiring has well-distributed facial features. The results of the current step derived solely using geometric information are guided by the minimum values of the constraint functions. The optical flow registration step draws inspiration from traditional manual facial markers techniques in the context of retopology. We use the optical flow to automatically constrain NICP registration to match facial texture information, thus obtaining more nonlinear facial topology results. For each raw face model, optical flow is computed through virtual camera rendering. Vertex projections onto the image coordinate system provide texture constraints, and the deformation map of vertices is updated during the iterative process.

2.3 Sequence Refinement and Filter

Due to factors such as camera coverage angle and headgear, the 3D facial model still contains many outliers. As shown in Fig. 1, these outliers are mainly concentrated in the headgear and occipital region. Since this region does not involve any expressions, and the motion of the skull has been decoupled and eliminated, we can use a weight mask to blend the target expression model with the neutral template model. Additionally, white noise exists in vertices due to imaging, random feature matching, and other processes. From a temporal perspective, this noise distribution is independent of the

facial position. We apply adaptive threshold wavelet denoising to the coordinate motion curves of each vertex in the sequence, thus obtaining a model sequence result with vertex-level granularity and stability.



Figure 3: The scan-to-mesh distance is visualized in heatmap.

3 RESULTS

Fig. 1 also illustrates extreme expressions results from the captured sequence. The proposed retopology method effectively handles expressions involving closed eyes, addressing the issue of incomplete eye closure that can occur with standard NICP methods due to smoothness constraints. Additionally, for regions such as lips, the fine-grained registration provided by optical flow results in more nuanced expressions. Moreover, the retopology method accurately preserves the geometric details of the original model. The error before and after retopology is shown in Fig. 3, with the majority of errors being below 0.5mm.

4 CONCLUSION

This paper introduces a 4D facial capture pipeline incorporating progressive retopology approach. The use of a high-speed camera array enables the system to capture facial movements at a frame rate up to 60Hz. The nonlinear deformations of the face are efficiently and accurately decoupled through a simple external device. The progressive retopology method aligns the raw model robustly and accurately. The pipeline maximizes the utilization of both geometric and texture information output by the capture system. The constraint of optical flow can be considered an equivalent alternative to traditional manual labeling. Experimental results demonstrate that our approach achieves precise and stable 4D model sequences, maintaining the personalized facial movement characteristics of different characters. This method holds promising applications in digital human capture and generation, deep learning data generation, and beyond, making a significant impact in various fields.

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